

```
from google.colab import files
uploaded = files.upload()

import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread(list(uploaded.keys())[0])
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

blurred = cv2.GaussianBlur(gray, (5, 5), 0)

mask = cv2.subtract(gray, blurred)

k = 1.5
sharpened = cv2.addWeighted(gray, 1.0, mask, k, 0)

plt.figure(figsize=(14,5))

plt.subplot(1,4,1)
plt.title("Original Image")
plt.imshow(gray, cmap='gray')
plt.axis('off')

plt.subplot(1,4,2)
plt.title("Blurred Image")
plt.imshow(blurred, cmap='gray')
plt.axis('off')

plt.subplot(1,4,3)
plt.title("Mask (Details)")
plt.imshow(mask, cmap='gray')
plt.axis('off')

plt.subplot(1,4,4)
plt.title("Sharpened Image (Unsharp Masking)")
plt.imshow(sharpened, cmap='gray')
plt.axis('off')

plt.show()
```

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Original Image



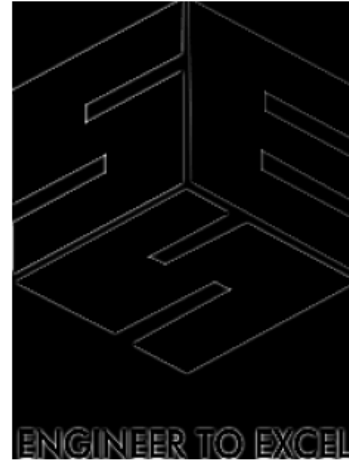
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Blurred Image



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Mask (Details)



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Sharpened Image (Unsharp Masking)



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